

Study (Title)	Year	Domain	VR system / HMD	Omnidirectional treadmills (ODTs) or locomotion interface	Methods	Measures	Main findings	Limitations
Influence of guidance signs on platform evacuation in suburban railway tunnel under smoke and obstacle environment	2025	Suburban Railway / Emergency Evacuation	HTC Vive	Omnidirectional treadmill (with sensor-equipped shoes)	VR experiment simulating smoke and obstacles on a narrow tunnel platform to compare No Guiding (NG), Wall-Guided (WG), and Central axis Guidance (CG) signs.	Forward movement speed, wall distance (lateral position), coefficient of variation (CV), and psychological stress (10-point scale).	• Guidance signs significantly reduced participants' mental stress under smoky conditions. • CG signs improved speed stability and reduced fluctuations when circumventing obstacles. • WG signs led participants to walk closer to walls, which was effective for obstacles on the wall side.	• Limited to a straight tunnel platform; curved tunnels or higher density smoke not yet studied. • Potential learning effects as participants performed multiple trials. • Heterogeneity among participants' digital literacy was not deeply analyzed.
Using Fully Immersive Virtual Reality to Teach Safe Pedestrian Skills to Autistic Children	2025	Pediatric Safety / Special Education	HTC Vive Pro 2	Infinadeck Omnidirectional Treadmill	Concurrent multiple probe design using backward chaining and schedule thinning to teach road crossing (clear, busy, and car-stopped conditions) in a Virtual Environment (VE).	Percentage of correct behavioral steps (core and vocal), number of trials with 100% core steps, hits/near misses, and social validity ratings.	• Three of four participants achieved 100% correct independent road crossings that generalized to the natural environment. • Gamification elements (tokens) promoted independent movement without instructor prompts. • Mastering verbal component skills did not automatically lead to safe physical crossing behavior without direct VR training.	• High equipment cost (approx. \$40,000\$ for treadmill) limits broad accessibility. • Small sample size (4 children) and lack of fully extended baselines in multiple probe design. • Technical difficulties (sensor loss, hardware failure) delayed study progression.
The Impact of Corridor Directional Configuration on Wayfinding Behavior during Fire Evacuation in Underground Spaces: An Empirical Study Based on Virtual Reality	2024	Underground Space Emergency Evacuation	HTC Vive Pro Eye	Virtuix Omni	VR experiment with 115 volunteers in two scenarios (real-mall based and abstract) with Y, T, $\perp$, and $\top$ shaped intersections; signage removed.	Movement trajectories (coordinates every 0.5s), choice percentages at intersections, decision-making time, and chi-square/logistic regression analysis.	• Significant preference for right-side corridors at Y-shaped (58.3%) and T-shaped (61.7%) intersections. • Strong preference for straight paths (~70%) over turning at $\perp$ and $\top$ intersections. • Individual factors (gender, professional background) did not significantly affect directional choices.	• Psychological and physiological responses in VR may differ from real fire stress. • Sample bias: primarily university students, lacking elderly or diverse mobility groups. • Short-term experiments cannot simulate long-term, large-scale multi-factor evacuation complexity.
A virtual reality study of behavioral sequences in residential fires	2021	Fire Safety / Emergency Evacuation	HTC Vive	Cyberith Virtualizer	Participants were exposed to virtual fire scenarios (with/without smoke alarms) and their behavioral sequences were recorded and compared to the Canter, Breaux, and Sime general model.	Behavioral sequences (interpret, prepare, act), transition matrices, decomposition diagrams, and qualitative ratings of realism and stress.	• Participants' behavior sequences in VR closely aligned with patterns predicted by real-world historical models. • VR effectively triggered realistic behaviors such as ducking under smoke and repeated fire-fighting attempts. • Half of the participants in the alarm scenario ignored or misinterpreted the sound.	• Small sample size (40 participants) prevented robust quantitative statistical comparisons. • Significant incidence of motion sickness (8 participants interrupted the study). • Difficulty in distinguishing between misinterpretation of cues versus simply not noticing them.

Design and Implementation of Coal Mine Flood Rescue Drill System Based on VR	2026	Coal Mine Safety / Flood Rescue	HTC Vive	VR universal treadmill (with infrared sensors)	Developed a comprehensive drill system using 3D scene reconstruction, dynamic fluid simulation for water inrush, and multi-role real-time collaborative interaction.	Step score (accuracy of 12 key steps) and quality score (Likert scale for proficiency, safety, and teamwork) evaluated by experts.	• VR system successfully restored complex underground flood characteristics including rising water levels and disordered ventilation. • High-fidelity simulation improved risk identification and practical proficiency in first aid and drainage operations. • Quantitative evaluation allowed for personalized training feedback and error analysis.	• Fluid simulation accuracy is simplified for real-time use and lacks complex multisource water coupling. • VR interaction currently lacks fine whole-body feedback (e.g., walking resistance in water). • Limited to a single disaster type (flood) without integrated compound disasters.
A Comprehensive Framework for Integrating Extended Reality into Lifecycle-Based Construction Safety Management	2025	Construction Safety Management	HTC Vive, Android smartphone (AR viewer)	Not in source	Conceptual and methodological study involving a systematic literature review, analytical categorization of implementation requirements, and framework development supported by illustrative examples.	Technical feasibility, internal consistency of the framework, and conceptual validation through simulated scenarios (e.g., fall risk analysis, PPE/CPE verification).	• Developed a structured lifecycle-based framework integrating BIM, Lean Construction, and Prevention through Design. • Demonstrated XR's capability for immersive walkthroughs, real-time hazard identification, and automated safety inspections. • Highlighted the importance of storytelling and gamification to enhance safety awareness.	• Framework not yet deployed in a complete, real-world project to measure long-term effectiveness. • Lack of massive field testing and quantitative safety indicators. • Potential for technical failure or misalignment with real-site conditions causing false confidence.
Exploring the Impact of Digital Cultural Heritage Experiences on Generation Z's eWOM Intention	2026	Digital Cultural Heritage / HCI	Multi-functional digital platform	Not in source	Questionnaire survey (342 responses) analyzed via PLS-SEM based on affordance theory and the S-O-R (Stimulus-Organism-Response) model.	Information affordance, interaction affordance, emotional value, cultural identity, and electronic word-of-mouth (eWOM) intention.	• Information and interaction experiences significantly enhance emotional engagement and cultural belonging. • Narrative appeal and cultural symbols strengthen cultural identity. • Higher emotional value and identity lead to increased willingness to share/recommend content online.	• Cross-sectional data prevents drawing strong causal inferences. • Sample skewed towards students, limiting generalizability. • Context-bound to a single cultural platform (Palace Museum).